Chapter 3
Vector Spaces

3.3 Linear Independence

spanning set $\Leftarrow$ $Ax=b$ consistent? $\forall b \in \mathbb{R}^m$

linearly independent set

$\Uparrow$

$Ax=0$ has only the trivial solution?

Linear Dependence

Definition:
The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in a vector space V are said to be **linearly dependent (线性相关)** if and only if there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n$ **,not all zero,** such that

$\exists \alpha_i \neq 0$

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n = 0$$

- Let $A = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n]$, then $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is linearly dependent if and only if the null space of A, $N(A)$, is NOT the zero subspace {**0**}.
- **Theorem**
  Nonzero vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly dependent if and only if at least one vector is a linear combination of the others.

solution set of $Ax=0$

① $v_1 \cdots v_n$ linearly dependent →

at least one vector can —— linear ——

$$\alpha_1 v_1 + \cdots + \alpha_i v_i + \cdots + \alpha_n v_n = 0$$

Since $v_1 \cdots v_n$ are linearly dependent.

$\alpha_1 \cdots \alpha_n$ are not all 0.

Assume $\alpha_i \neq 0$.

$$v_i = -\frac{1}{\alpha_i}(\alpha_1 v_1 + \cdots + \alpha_{i-1} v_{i-1} + \alpha_{i+1} v_{i+1} + \cdots + \alpha_n v_n)$$

$$= -\frac{\alpha_1}{\alpha_i} v_1 - \cdots - \frac{\alpha_{i-1}}{\alpha_i} v_{i-1} - \frac{\alpha_{i+1}}{\alpha_i} v_{i+1} - \cdots - \frac{\alpha_n}{\alpha_i} v_n$$

$\downarrow$

v_i can be written as the linear combination of the others.

(2) at least one ... linear combination of others

$\rightarrow$ $v_1 \cdots v_n$ are linearly dependent.

$$v_i = \alpha_1 v_1 + \cdots + \alpha_{i-1} v_{i-1} + \alpha_{i+1} v_{i+1} + \cdots + \alpha_n v_n$$

$\Downarrow$

$$\alpha_1 v_1 + \cdots + \alpha_{i-1} v_{i-1} - v_i + \alpha_{i+1} v_{i+1} + \cdots + \alpha_n v_n = 0$$

$\alpha_i = -1$

$\alpha_1 \cdots \alpha_n$ are not all zero $\Rightarrow$ $v_1 \cdots v_n$ linearly dependent.

Eg. $\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}, \begin{pmatrix}1\\2\end{pmatrix}$ are linearly dependent.

① $\begin{pmatrix}1\\2\end{pmatrix} = 1\times\begin{pmatrix}1\\0\end{pmatrix} + 2\begin{pmatrix}0\\1\end{pmatrix}$

② $Ax=0$

$$\left[\begin{array}{ccc|c}1 & 0 & 1 & 0\\ 0 & 1 & 2 & 0\end{array}\right]$$

$Ax=0$ has a free variable

↓

it has a non-trivial solution

Def. $V = R^n$

$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0$

$\alpha_1, \cdots, \alpha_n$ not all zero?

Let $A = [v_1 \cdots v_n]_{n \times n}$ Let $x = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}_{n \times 1}$

$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0$

$\Updownarrow$

$Ax = 0$ homogeneous system

$v_1 \cdots v_n$ linearly dependent

$\Updownarrow$

$Ax = 0$ has a non-trivial solution.

Linear Independence

Definition:

The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in a vector space V are said to be **linearly independent (线性无关)** if

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n = 0$$

implies that all the scalers $\alpha_1, \alpha_2, \ldots, \alpha_n$ **must equal to** 0.

- Let $A = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n]$, then $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is linearly independent if and only if the null space of A IS the zero subspace: $N(A) = \{\mathbf{0}\}$.
- None of the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ can be written as a linear combination of the others.

$\begin{pmatrix}1\\0\end{pmatrix}$ $\begin{pmatrix}0\\1\end{pmatrix}$ linearly independent

$\left[\begin{array}{cc|c}1 & 0 & 0\\ 0 & 1 & 0\end{array}\right]$

$c_1v_1 + \cdots + c_nv_n = 0$

- $c_1 \cdots c_n$ all 0 → linearly independent.
- $c_1 \cdots c_n$ not all 0 → linearly dependent.

If $V = \mathbb{R}^n$

$Ax = 0$

- has the unique trivial solution
- has a non-trivial solution

$N(A) = \{0\}$ zero-subspace.

$Ax = b$ is always consistent for $\forall b \in \mathbb{R}^n$

Spanning set

Summary. If $V = \mathbb{R}^n$

Let $A = [v_1 \cdots v_n]$ ($m \times n$)

① $Ax = 0$ has the unique trivial solution

⇕

$N(A) = \{0\}$ (zero-subspace.)

⇕

The columns of A are linearly independent.

⇕

If $A: n \times n$, A is nonsingular.

$\Updownarrow$

If $A: n \times n$ $\det(A) \neq 0$

$\Updownarrow$

If $A: n \times n$ $Ax = b$ has the unique solution for $\forall b \in \mathbb{R}^n$

$\Updownarrow$

If $A: n \times n$

The columns of A form a spanning set of $\mathbb{R}^n$

(2) $Ax = 0$ has a nontrivial solution

$\Updownarrow$

$N(A) \neq \{0\}$

$\Updownarrow$

The columns of A are linearly dependent.

$\Updownarrow$

If $A: n \times n$ A is singular.

$\Updownarrow$

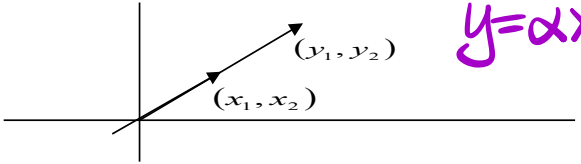
If $A: n \times n$ $\det(A) = 0$

Quick Example 1

- When is a set of two nonzero vectors linearly dependent?

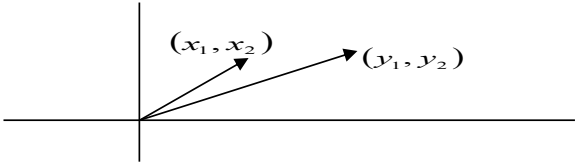
R^2

When one vector can be written as a scaler multiple of the other. Geometrically, if the two vectors are placed at the origin, they will lie along the same line.



$y = \alpha x$

x and **y** linearly dependent



x and **y** linearly independent

- Is a set containing the zero vector linearly dependent or linearly independent?

Linearly dependent. Let the scalers of all other vectors be zero, and the scaler of the zero vector be nonzero.

$0 = 0v_1 + 0v_2 + \cdots + 0v_{n-1}$

Example 2

Are the vectors $\mathbf{v}_1 = \begin{bmatrix}1\\0\\1\\2\end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix}0\\1\\1\\2\end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix}1\\1\\1\\3\end{bmatrix}$ in $\mathbb{R}^4$ linearly dependent or independent?

$Ax=0$ — unique trivial → no free variable.

non-trivial → there is a free variable

r.r.e.f

$$\left[\begin{array}{ccc|c}1&0&1&0\\0&1&1&0\\1&1&1&0\\2&2&3&0\end{array}\right] \xrightarrow[R_4-2R_1]{R_3-R_1} \left[\begin{array}{ccc|c}1&0&1&0\\0&1&1&0\\0&1&0&0\\0&2&1&0\end{array}\right] \xrightarrow[R_4-2R_2]{R_3-R_2} \left[\begin{array}{ccc|c}1&0&1&0\\0&1&1&0\\0&0&-1&0\\0&0&-1&0\end{array}\right]$$

$$\xrightarrow{R_4-R_3} \xrightarrow{-R_3} \left[\begin{array}{ccc|c}1&0&1&0\\0&1&1&0\\0&0&1&0\\0&0&0&0\end{array}\right]$$

unique trivial

no free variable

Example 2

Are the vectors $\mathbf{v}_1 = \begin{bmatrix}1\\0\\1\\2\end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix}0\\1\\1\\2\end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix}1\\1\\1\\3\end{bmatrix}$ in $\mathbb{R}^4$ linearly dependent or independent?

Solution If $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3$=0, solve the homogenous linear system

$$\begin{aligned}\alpha_1 + \alpha_3 &= 0\\ \alpha_2 + \alpha_3 &= 0\\ \alpha_1 + \alpha_2 + \alpha_3 &= 0\\ 2\alpha_1 + 2\alpha_2 + 3\alpha_3 &= 0\end{aligned}$$

By Gauss-Jordan reduction:

$$\left[\begin{array}{ccc|c}1 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0\end{array}\right]$$

The only solution is $\alpha_1 = \alpha_2 = \alpha_3 = 0$.
Therefore $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ are linearly independent.

Example 3

Are the vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix}$ and $\mathbf{v}_4 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$ in $\mathbb{R}^3$ linearly dependent or independent?

$$A = \begin{bmatrix} 1 & 1 & -3 & 2 \\ 2 & -2 & 2 & 0 \\ -1 & 1 & -1 & 0 \end{bmatrix} \quad 3 \times 4$$

$m \times n$ homogeneous system with $m < n$,

it has infinitely many solutions.

$\downarrow$

linearly dependent.

Example 3

Are the vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix}$ and $\mathbf{v}_4 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$ in $\mathbb{R}^3$ linearly dependent or independent?

Solution

If $\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \alpha_3\mathbf{v}_3 + \alpha_4\mathbf{v}_4 = 0$, solve the homogenous linear system

$$\begin{aligned} \alpha_1 + \alpha_2 - 3\alpha_3 + 2\alpha_4 &= 0 \\ 2\alpha_1 - 2\alpha_2 + 2\alpha_3 &= 0 \\ -\alpha_1 + \alpha_2 - \alpha_3 &= 0 \end{aligned}$$

By Gauss-Jordan reduction:

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

The solutions of the system are $[\alpha_3 - \alpha_4, 2\alpha_3 - \alpha_4, \alpha_3, \alpha_4]$, so there are nontrivial solutions, therefore the vectors are linearly dependent.

Example 4

Are the vectors in P_3 (polynomial functions)

$$p_1(t) = t^2 + 4t + 14$$
$$p_2(t) = 2t^2 + t$$

linearly dependent or independent?

P_n: contains all polynomials with degree less than n

Definition: $\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = 0$

$\alpha_1 \cdots \alpha_n$ all 0 or not all 0?

$$\alpha_1 p_1(t) + \alpha_2 p_2(t) = 0$$

$$\alpha_1(t^2+4t+14) + \alpha_2(2t^2+t) = 0t^2+0t+0$$

$$(\alpha_1+2\alpha_2)t^2+(4\alpha_1+\alpha_2)t+14\alpha_1=0t^2+0t+0$$

$$\begin{cases}\alpha_1+2\alpha_2=0\\ 4\alpha_1+\alpha_2=0\\ 14\alpha_1=0\end{cases}\Rightarrow\begin{cases}\alpha_1=0\\ \alpha_2=0\end{cases}\Rightarrow$$

$p_1(t)$ and $p_2(t)$ are linearly independent

Example 4

Are the vectors in P_3 (polynomial functions)

$$p_1(t) = t^2 + 4t + 14$$
$$p_2(t) = 2t^2 + t$$

linearly dependent or independent?

Solution

To test whether $p_1(t)$ and $p_2(t)$ are linearly independent, set

$$\alpha_1 p_1(t) + \alpha_2 p_2(t) = 0t^2 + 0t + 0$$

We get $(\alpha_1 + 2\alpha_2)t^2 + (4\alpha_1 + \alpha_2)t + 14\alpha_1 = 0$

Equating coefficient lead to the system

$$\alpha_1 + 2\alpha_2 = 0$$
$$4\alpha_1 + \alpha_2 = 0$$
$$14\alpha_1 = 0$$

The only solution is $\alpha_1 = 0, \alpha_2 = 0$.

Therefore $p_1(t)$ and $p_2(t)$ are linearly independent.

Theorem

Theorem 3.3.1

Let $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ be n vectors in $\mathbb{R}^n$ and let $X = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n]$. The vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ will be linearly dependent if and only if X is singular.

Warning: This theorem only works with n vectors in $\mathbb{R}^n$ (square matrix case)

Proof The equation

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \cdots + c_n\mathbf{x}_n = \mathbf{0}$$

can be rewritten as a matrix equation

$$X\mathbf{c} = \mathbf{0}$$

This equation will have a nontrivial solution if and only if X is singular. Thus, $\mathbf{x}_1, \ldots, \mathbf{x}_n$ will be linearly dependent if and only if X is singular. ■

We can use Theorem 3.3.1 to test whether n vectors are linearly independent in $\mathbb{R}^n$. Simply form a matrix X whose columns are the vectors being tested. To determine whether X is singular, calculate the value of $\det(X)$. If $\det(X) = 0$, the vectors are linearly dependent. If $\det(X) \neq 0$, the vectors are linearly independent.

Theorem 3.3.2

Let $\mathbf{v}_1, \ldots, \mathbf{v}_n$ be vectors in a vector space V. A vector $\mathbf{v} \in \text{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_n)$ can be written *uniquely* as a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_n$ if and only if $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are linearly independent.

$\forall v \in \text{Span}\{v_1 \cdots v_n\}$

$\exists \alpha_1 \cdots \alpha_n$ such that $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$

(1) unique combination $\rightarrow$ $v_1 \cdots v_n$ are linearly independent

Proof of contradiction

Assume. $v_1 \cdots v_n$ are linearly dependent.

$\exists\ c_1 \cdots c_n$ not all zero such that

$c_1 v_1 + \cdots + c_n v_n = 0$ ①

We know $0 \in \text{Span}\{v_1, \cdots, v_n\}$.

$0 = 0v_1 + 0v_2 + \cdots + 0v_n$ ②

$0 \in \text{Span}\{v_1 \cdots v_n\}$

$\begin{cases} 0 = c_1 v_1 + \cdots + c_n v_n & c_1 \cdots c_n \text{ not all zero} \\ 0 = 0v_1 + \cdots + 0v_n \end{cases}$

there are two different linear combinations.

Contradicts with unique combination

$\Rightarrow$ $v_1 \cdots v_n$ are linearly independent

② $v_1 \cdots v_n$ linearly independent $\rightarrow$ unique linear combination

Proof of contradiction.

Assume $\forall v \in \text{Span}\{v_1 \cdots v_n\}$, there exists two different linear combinations of v.

① $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$

② $v = \beta_1 v_1 + \cdots + \beta_n v_n$

① − ② $\Rightarrow \quad 0 = (\alpha_1 - \beta_1) v_1 + \cdots + (\alpha_n - \beta_n) v_n$

Since $v_1 \cdots v_n$ are linearly independent.

$\alpha_1 - \beta_1 = 0$ & $\cdots$ & $\alpha_n - \beta_n = 0$

$\Downarrow$

$\alpha_1 = \beta_1$ & $\cdots$ & $\alpha_n = \beta_n$

Contradicts with the assumption of two different combinations.